PostgreSQL & pgAdmin Setup Guide

Official Step-by-Step Installation & Configuration Walkthrough for Windows & macOS

Overview: This guide provides standard installation steps for setting up PostgreSQL (Database Engine) along with `pgAdmin 4` (GUI Management Tool), following standard industry practices and course recommendations.

1. WINDOWS INSTALLATION STEPS

STEP 1 Download the Installer

- Navigate to the official EnterpriseDB / PostgreSQL website download page.
- Select the Windows x86-64 installer for the latest stable version (e.g., PostgreSQL 15 or 16).

STEP 2 Run the Setup Wizard

- Launch the downloaded `.exe` file as Administrator.
- On the **Select Components** screen, keep all default options checked:
  - `PostgreSQL Server` (Core database engine)
  - `pgAdmin 4` (Graphical interface tool)
  - `Stack Builder` (Optional drivers tool)
  - `Command Line Tools` (psql, pg_dump)

STEP 3 Set Superuser Password & Port

- **Password Setup:** Assign a password for the default database superuser (`postgres`). Record this password carefully; it is required for all database connections.
- **Port Selection:** Leave the default port set to `5432` unless specifically occupied by another service.

2. MACOS INSTALLATION STEPS

METHOD A EDB Graphical Installer (Recommended)

1. Download the macOS `.dmg` installer package from the official PostgreSQL portal.
2. Open the installer package and follow the setup prompts similar to Windows.
3. Set the superuser (`postgres`) master password and retain default port `5432` .

METHOD B **Homebrew CLI Package Manager**

For developer setups using Homebrew terminal commands:

- Install PostgreSQL engine: `brew install postgresql`
- Start background service: `brew services start postgresql`
- Install pgAdmin 4 application: `brew install --cask pgadmin4`

3. INITIAL SETUP & PGADMIN ENVIRONMENT VERIFICATION

STEP 4 **Launching pgAdmin & Query Tool**

1. Open **pgAdmin 4** from your operating system application menu.
2. Set a Master Password for pgAdmin upon first launch.
3. Expand `Servers` in the left sidebar and select `PostgreSQL`. Enter the `postgres` user password created during setup.
4. Right-click on the default database (`postgres`) and select **Query Tool** to open the SQL editor panel.